

Annexure

Skill-Based Resume Ranking System with Explainable AI

CSE-414: Machine Learning and Deep Learning Lab

Department of Computer Science and Engineering, University of Asia Pacific

Tanjina Akter Nisa (22201086) Md. Daudul Islam (22201088) Saswata Ghosal
(22201090)

Appendix A: CEP Mapping

A.1 Knowledge Profile (WK)

WK Code	Description	How Addressed in Project	Report Section
WK1	Natural sciences applicable to the discipline.	Mathematics of cosine similarity computation, probability theory in classification, and weighted arithmetic in the hybrid scoring formula.	Section III (F)
WK2	Mathematics, numerical analysis, data analysis, statistics, and formal aspects of computer science.	TF-IDF vectorisation, label encoding, stratified 80/20 train-validation split, accuracy, precision, recall, and F1 score computation, and percentile-based candidate grouping.	Sec. II (B), IV (B)
WK4	Engineering specialist knowledge at the forefront of the discipline.	Transformer architecture (BERT family), fine-tuning of pre-trained models (DistilBERT), sentence embedding generation (SBERT), and NLP pipeline design.	Sec. III (D, E)
WK6	Knowledge of engineering practice in the discipline.	Industry-standard tools: Python, HuggingFace Transformers, scikit-learn, pdfplumber, Streamlit, pandas, and matplotlib.	Section IV (A)
WK8	Engagement with current research literature.	Application of BERT [1], DistilBERT [2], and Sentence-BERT [3] from current NLP literature. Critical evaluation of model tradeoffs.	Sec. III, V

A.2 Complex Engineering Problem (WP)

WP Code	Characteristic	How Addressed in Project	Report Section
WP1	Depth of knowledge required. Cannot be resolved without in-depth engineering knowledge at WK3-WK8 level.	System design required deep ML and DL knowledge: transformer architecture selection and fine-tuning (DistilBERT), sentence embedding theory (SBERT), hybrid scoring design, and NLP pipeline engineering.	Section III
WP2	Wide-ranging or conflicting technical and non-technical requirements.	Technical: balancing model accuracy with inference speed (ML vs. DL trade-off). Non-technical: ethical concerns of HR automation, bias reduction in hiring, and data privacy of candidate information.	Sec. I, V
WP3	No obvious solution; requires abstract thinking, creativity, and originality.	No standard solution exists for combining ML classification, DL semantic matching, and rule-based extraction in a single explainable HR screening pipeline. The hybrid scoring formula required original design.	Section III
WP4	Infrequently encountered issues or novel problems.	Applying transformer-based NLP (BERT, SBERT) to automated HR resume screening and ranking is a novel application with limited prior standardised approaches.	Section I

A.3 Complex Engineering Activities (EA)

EA Code	Characteristic	How Addressed in Project	Report Section
EA1	Involve the use of diverse resources including data, technologies, and analytical software.	Utilised Kaggle dataset (962 resumes), HuggingFace pre-trained transformer models, scikit-learn ML library, pdfplumber for PDF processing, Streamlit for web interface, and matplotlib for visualisation.	Sec. II, IV

Continued...

(Continued from previous page)

EA Code	Characteristic	How Addressed	Section
EA2	Require optimal resolution of interactions between technical and engineering issues.	Systematic ML vs. DL tradeoff analysis: Logistic Regression and Random Forest evaluated against DistilBERT to justify architectural decisions. Speed vs. accuracy tradeoffs were analysed and documented.	Section V (B)
EA3	Involve creative use of engineering principles and research-based knowledge.	Innovative combination of SBERT semantic matching, DistilBERT classification, and rule-based scoring in a unified pipeline. Explainable AI applied to HR recruitment — a novel engineering contribution built on current NLP research.	Section III

Appendix B: Team Member Contribution

Team Member	Contribution
Tanjina Akter Nisa (22201086)	• Dataset collection and inspection from Kaggle. • Data cleaning, null removal, and preprocessing pipeline design. • TF-IDF vectorisation and label encoding implementation. • Logistic Regression model training, hyperparameter tuning, and evaluation. • Computation of precision, recall, F1 score, and classification report. • Contribution to result analysis, interpretation, and presentation slides.
Md. Daudul Islam (22201088)	• DistilBERT fine-tuning pipeline design and training on resume dataset (3 epochs). • SBERT integration for semantic similarity scoring using cosine distance. • Hybrid scoring formula design (SBERT 40%, Skills 30%, Experience 15%, Education 15%). • PDF text extraction module using pdfplumber with regex-based cleaning. • Skill, experience, and education extraction using regex and fuzzy matching. • Project report writing (Sections I–VI) and reference management.
Saswata Ghosal (22201090)	• Random Forest model training, hyperparameter configuration, and evaluation. • ML vs. DL comparative analysis and visualisation (bar charts). • Streamlit web application design and development (Login, Scan, Output pages). • Advanced analytics: percentile grouping, group insights, score breakdown charts. • CEP mapping (Knowledge Profile, Complex Problem, Engineering Activities). • Presentation slide preparation, proofreading, and demo preparation.

Appendix C: Additional Results

C.1 Full Classification Report — Logistic Regression

Category	Precision	Recall	F1	Support
Advocate	1.00	1.00	1.00	4
Arts	1.00	1.00	1.00	7
Automation Testing	0.83	1.00	0.91	5
Blockchain	1.00	1.00	1.00	8
Business Analyst	1.00	1.00	1.00	6
Civil Engineer	1.00	1.00	1.00	5
Data Science	1.00	1.00	1.00	8
Database	1.00	1.00	1.00	7
DevOps Engineer	1.00	0.91	0.95	11
DotNet Developer	1.00	1.00	1.00	5
ETL Developer	1.00	1.00	1.00	8
Electrical Engineering	1.00	1.00	1.00	6
HR	1.00	1.00	1.00	9
Hadoop	1.00	1.00	1.00	8
Health and Fitness	1.00	1.00	1.00	6
Java Developer	1.00	1.00	1.00	17
Mechanical Engineer	1.00	1.00	1.00	8
Network Security Eng.	1.00	1.00	1.00	5
Operations Manager	1.00	1.00	1.00	8
PMO	1.00	1.00	1.00	6
Python Developer	1.00	1.00	1.00	10
SAP Developer	1.00	1.00	1.00	5
Sales	1.00	1.00	1.00	8
Testing	1.00	1.00	1.00	14
Web Designing	1.00	1.00	1.00	9
Accuracy	99.48%			193
Weighted Avg	1.00	0.99	0.99	193

C.2 Full Classification Report — DistilBERT

Category	Precision	Recall	F1	Support
Advocate	1.00	1.00	1.00	4
Arts	1.00	1.00	1.00	7
Automation Testing	1.00	1.00	1.00	5
Blockchain	1.00	1.00	1.00	8
Business Analyst	1.00	1.00	1.00	6
Civil Engineer	1.00	1.00	1.00	5
Data Science	1.00	1.00	1.00	8
Database	1.00	1.00	1.00	7
DevOps Engineer	1.00	0.91	0.95	11
DotNet Developer	1.00	1.00	1.00	5
ETL Developer	1.00	1.00	1.00	8
Electrical Engineering	1.00	1.00	1.00	6
HR	1.00	1.00	1.00	9
Hadoop	1.00	1.00	1.00	8
Health and Fitness	1.00	1.00	1.00	6
Java Developer	1.00	1.00	1.00	17
Mechanical Engineer	1.00	1.00	1.00	8
Network Security Eng.	1.00	1.00	1.00	5
Operations Manager	1.00	1.00	1.00	8
PMO	1.00	1.00	1.00	6
Python Developer	1.00	1.00	1.00	10
SAP Developer	1.00	1.00	1.00	5
Sales	1.00	1.00	1.00	8
Testing	0.93	1.00	0.97	14
Web Designing	1.00	1.00	1.00	9
Accuracy	99.00%			193
Weighted Avg	1.00	0.99	0.99	193

C.3 Hybrid Scoring Formula — Component Weights Justification

Component	Weight	Justification	Calculation
SBERT Semantic Similarity	40%	Most comprehensive relevance measure; captures meaning beyond keywords. Highest weight reflects ability to evaluate overall CV–job alignment.	Cosine similarity between CV and JD embeddings
Skill Match	30%	Primary hiring criterion; directly measures technical fit. Second highest weight.	<code>matched_skills / total_required_skills</code>
Experience	15%	Important but not always clearly stated in CVs; regex extraction has limitations.	<code>min(CV_years / req_years, 1.0)</code>
Education	15%	Baseline requirement; equal weight to experience.	1.0 if met, 0.5 if one below, 0.0 otherwise

References

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proc. NAACL-HLT*, Minneapolis, MN, USA, 2019, pp. 4171–4186.
- [2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [3] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proc. EMNLP-IJCNLP*, Hong Kong, China, 2019, pp. 3982–3992.